WHICH ASPECT, WHICH OPINION?

Aspect-based sentiment analysis cast as BIO sequence labeling: every token gets one tag.

I	love	Windows 7	which	is	a	vast	improvement	over	Vista
O	O	B-/I-ASP_POS	O	O	O	B-OPINION	O	O	B-ASP_NEG

Aspects and polarities are gold SemEval labels; **OPINION** is rule-based (adjectives/adverbs only), so the verb “love” stays O.

4,707

SemEval-2014 laptop + restaurant review sentences

3,765 / 942

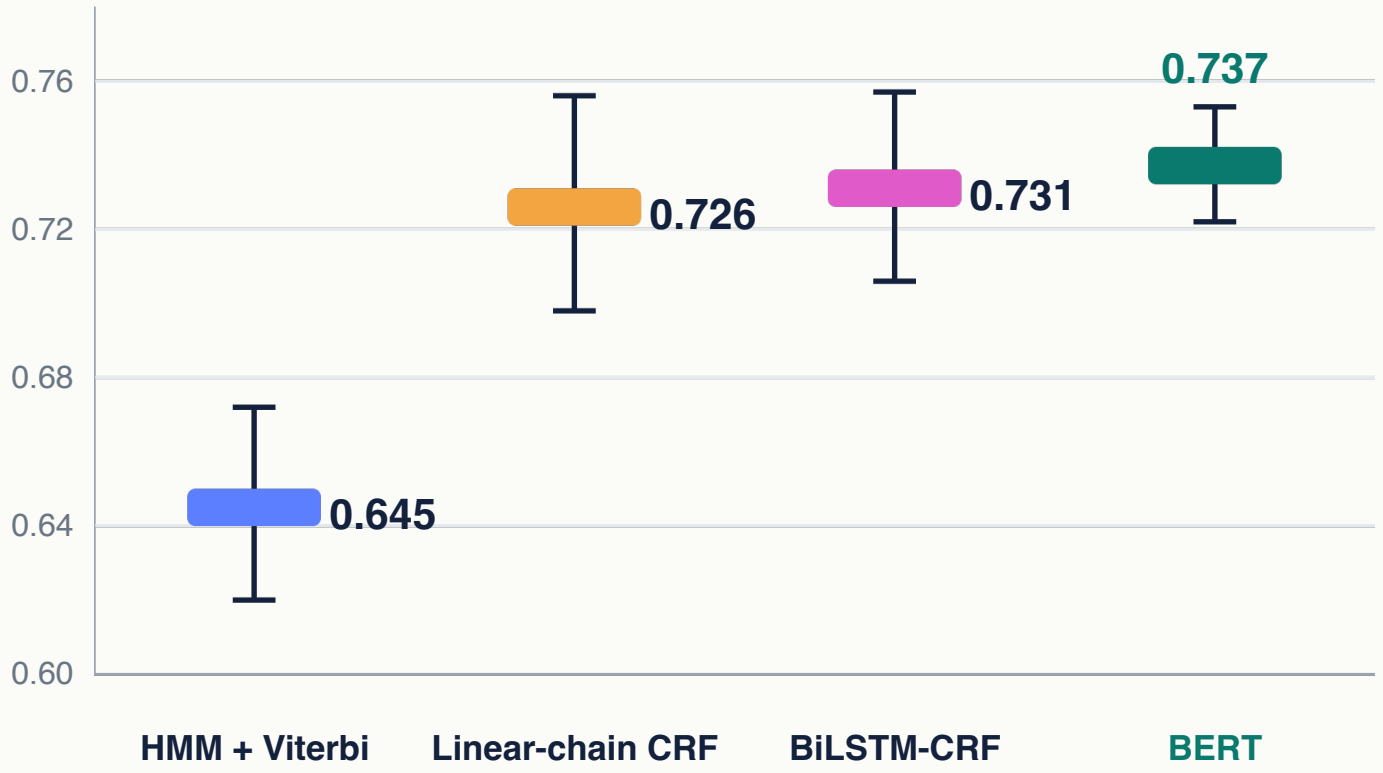
train / test · 10% of train held out for validation

9 tags

3 polarity-aware aspect types + OPINION, in B/I form, plus O

FOUR TAGGERS, RANKED BY F1

Same split, scored with span-level exact match (seqeval, micro-F1).



HOW TO READ

Box = micro-F1 · line = precision ↔ recall (one seed, no variance).

BERT
contextual · P .722 R .753

BiLSTM-CRF
learned · P .757 R .706

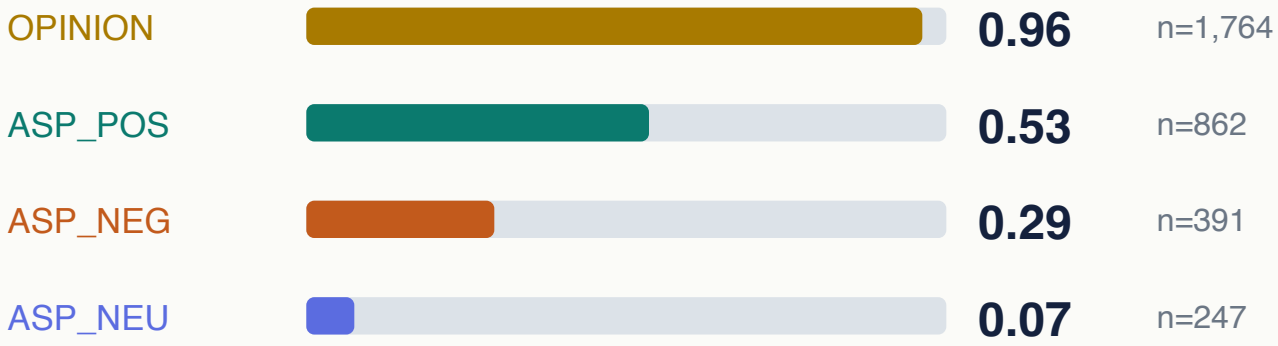
Linear-chain CRF
hand features · P .756 R .698

HMM + Viterbi
generative · P .672 R .620

MICRO-F1 HIDES THE HARD PART

OPINION labels come from a POS rule (JJ/RB), so the model largely re-learns the tagger.

PER-CLASS F1 · BILSTM-CRF



Macro-F1 **0.47** vs. micro-F1 **0.73**: rare polarities pay the price.

SENTENCE ERROR RATE

Contrast	**0.84** vs 0.70
Multiple aspects	**0.82** vs 0.64
Negation	**0.80** vs 0.69

“The battery life has **not decreased** … I’m thrilled”: gold POS, predicted NEG.

FROM TAGS TO STRUCTURED KNOWLEDGE

Predicted spans become records a system can query.

STEP 1
BIO spans per sentence

→

STEP 2
(aspect, opinion, polarity) triplets

→

STEP 3
Aspect search + graph

PAIRING: NEAREST OPINION SPAN

Works (food, absolutely delicious, POS)

Works (keyboard, good, POS)

Fails (battery life, not, NEG)

Fails (sandwiches, 7, NEG)

LIMITATION → NEXT STEP

Heuristic opinion labels → gold triplet annotations

Distance-based pairing → dependency-based linking

One seed, κ on a demo set → repeated runs, real double annotation